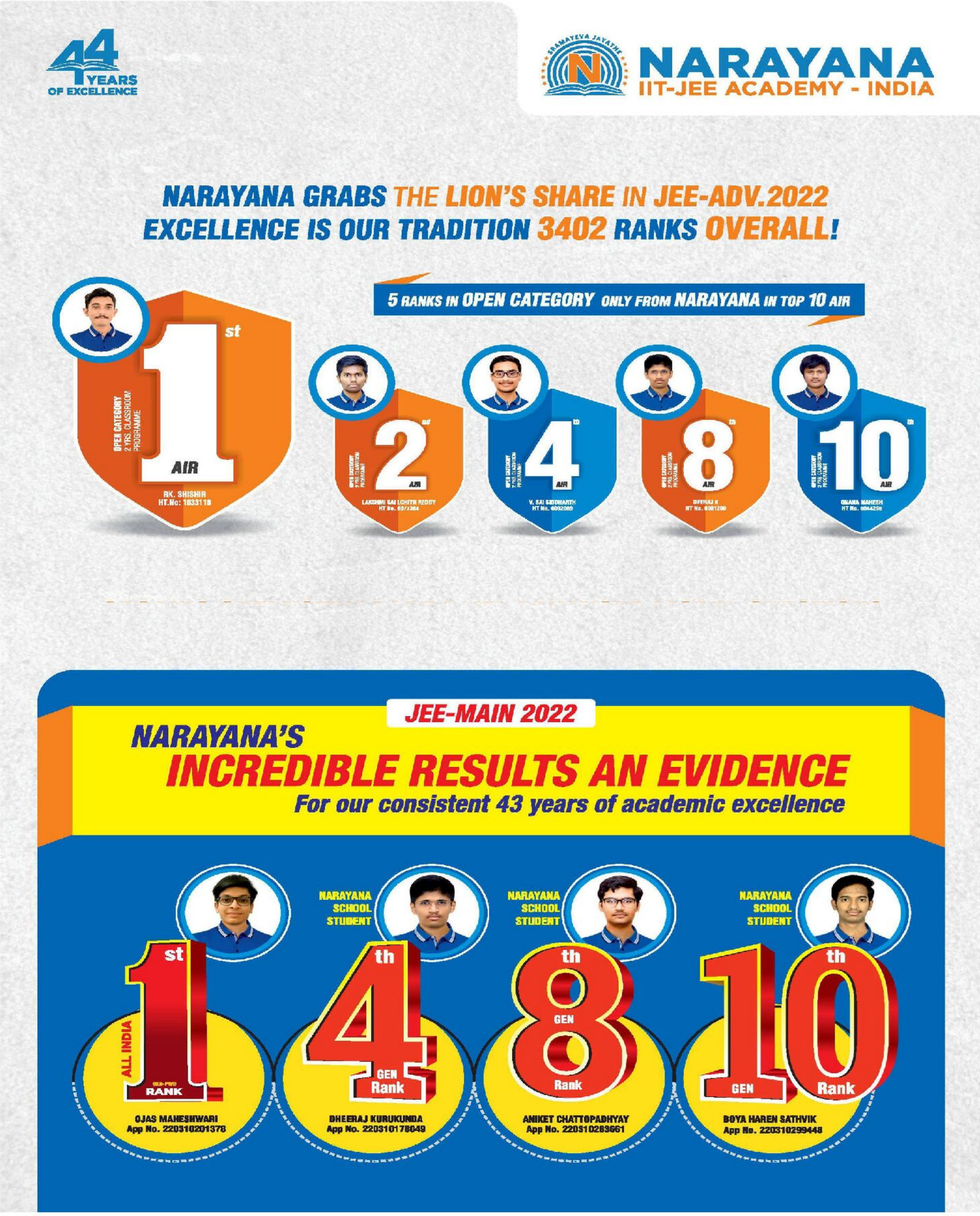


CHEMISTRY

INORGANIC CHEMISTRY PART-I(JR)

JEE MAIN IMPORTANT QUESTIONS

INORGANIC CHEMISTRY PART-I(JR)

Hydrogen & its Compounds

01. Identify the reaction which does not liberate hydrogen :

(1) Reaction of zinc with aqueous alkali.

(2) Electrolysis of acidified water using Pt electrodes.

(3) Allowing a solution of sodium in liquid ammonia to stand.

(4) Reaction of lithium hydride with B_2H_6

02. Which of the following statement is incorrect regarding the hydrides ?

1) Hydrides of Be and Mg are polymeric in structure

2) Stannane is an electron precise hydride

3) Diborane act as Lewis acid

4) Interstitial hydrides are formed by many s – block and p – block elements

03. How many hydrogen bonded water molecule(s) are associated in $CuSO_4.5H_2O(s)$?

04. The wrong statement is

1) By using Synthetic resin method all cations and anions including H^+ and OH^- are removed from water.

2) The exhausted permutite is regenerated by passing aqueous NaCl solution through it

3) Molarity of 10 "vol" H_2O_2 is 0.893M

4) 20ml of perhydrol on heating decomposes to give approximately 2 litres of O_2 at STP.

05. In 100 ml sample of hard water, 100 ml of (N/50) Na_2CO_3 was added and the mixture was boiled and filtered. The filtrate was neutralised with 60 ml of (N/50) HCl. Calculate the degree of permanent hardness of water (in ppm of $CaCO_3$). (sp. Gravity of hard water = 1)

06. Identify the incorrect statement regarding heavy water :

(1) It reacts with Al_4C_3 to produce CD_4 and $Al(OD)_3$.

(2) It is used as a coolant in nuclear reactors.

(3) It reacts with CaC_2 to produce C_2D_2 and $Ca(OD)_2$

(4) It reacts with SO_3 to form deuterated sulphuric acid (D_2SO_4)

07. H_2O_2 is used to restore the colour of old oil paintings. In the reaction responsible for this use, the change in oxidation number of Sulphur atom is

08. With which of the following H_2O_2 depicts its reducing property?

1) $K_4\left[Fe(CN)_6\right]$ in acidic medium.

2) $Cr(OH)_3$ in the presence of $NaOH$

3) PbS 4) $KMnO_4$ in the presence of H_2SO_4

09. Decreasing order of the hydrogen bonding in following forms of water is correctly represented by

A) Liquid water B) Ice C) Impure water

1) A = B > C 2) B > A > C 3) C > B > A 4) A > B > C

10. A) H_2O_2 is used synthesis of cephalosporin.

B) H_2O_2 is used in restoration of aerobic conditions to sewage wastes.

C) H_2O_2 is used in to manufacture sodium perborate and per-carbonate, which are used in high quality detergents

D) H_2O_2 is used as hair bleach and as a mild disinfectant.

The correct statements are

1) A, B only 2) C only 3) A,B,C,D 4) B,C,D only

11. A) Tanks of metal alloy like $NaNi_5, Ti-TiH_2, Mg-MgH_2$ etc. are in use for storage f dihydrogen in small quantities.

B) A cylinder of compressed dihydrogen weighs about 30 times as much as a tank of petrol containing the same amount of energy.

C) Hydrogen is used as a rocket fuel in space research.

D) Hydrogen is an environmental friendly fuel.

The correct statements are

1) A, B, C, D 2) B, C only 3) A, B, C only 4) A,C,D only

12. Match List I with List II

LIST-I	LIST-II
A) Cobalt catalyst	I) $(H_2 + Cl_2)$ production
B) Iron catalyst	II) Water gas production
C) Nickel catalyst	III) Ammonia production
D) Brine solution	IV) Methanol production

Choose the **correct** answer from the options given below:

1) $A-IV, B-I, C-II, D-III$ 2) $A-IV, B-III, C-I, D-II$

3) $A-II, B-III, C-IV, D-I$ 4) $A-IV, B-III, C-II, D-I$

13. In how many of the following reactions H_2 is obtained as one of the product ?

(i) $Al + NaOH + H_2O \rightarrow$

(ii) $Fe(heated) + H_2O(steam) \rightarrow$

(iii) $C(hot) + H_2O(steam) \xrightarrow{1270K}$

(iv) $CaH_2 + H_2O \rightarrow$

(v) NaH (fused) $\xrightarrow{electrolysis}$

(vi) Electrolysis of brine solution

(vii) Electrolysis of warm aq Barium hydroxide solution between nickel electrodes

(viii) $CH_4(g) + H_2O(g) \xrightarrow[Ni]{1270K}$

14. The amount(in grams) of H_2 liberated when 32.5 gm of Zn reacts with excess aq NaOH is ……..

15. **How many of the following properties are more for D_2O than H_2O?**

Melting point, Boiling point, Enthalpy of vaporisation, Density (at 25^0C), Dielectric constant, degree of dissociation

PERIODIC CLASSIFICATION OF ELEMENTS

16. Group number and electronic configuration of element of Atomic number 120 are

1) $2,[Uuh]8s^2$ 2) $2,[Uuo]8s^2$ 3) $3,[Uus]8s^2 8p^1$ 4) $3,[Uup]8s^2 8p^1$

17. The outer electron configuration of "Gd" (atomic no: 64) is

1. $4f^4\ 5d^4\ 6s^2$ 2. $4f^7\ 5d^1\ 6s^2$ 3. $4f^3\ 5d^5\ 6s^2$ 4. $4f^8\ 5d^0\ 6s^2$

18. Which of the following arrangements represents the increasing order (smallest to largest) of ionic radii of the given species O^{2-}, S^{2-}, N^{3-}, P^{3-},

1) $O^{2-} < N^{3-} < S^{2-} < P^{3-}$ 2) $O^{2-} < P^{3-} < N^{3-} < S^{2-}$

3) $N^{3-} < O^{2-} < P^{3-} < S^{2-}$ 4) $N^{3-} < S^{2-} < O^{2-} < P^{3-}$

19. Similarity in the radius of Zr and Hf is explained on the basis of

1) Lanthanide contraction 2) Inert pair effect

3) Same outer shell configuration 4) High effective nuclear charge

20. First Ionization enthalpies of few third period elements are 737, 496, 786 and 590 KJ/mole respectively. Then the elements respectively are

1) Si, Al, Mg, Na 2) Mg, Al, Si, Na 3) Mg, Na, Si, Al 4) Si, Na, Al, Mg

21. Consider the following ionization steps:

$M(g) \rightarrow M^+(g) + e^-; \Delta H = +100\ eV$; $M(g) \rightarrow M^{2+}(g) + 2e^-; \Delta H = +250\ eV$

Select the incorrect statement(s):

1) $I.E_1$ of $M(g)$ is 100 eV 2) $I.E_1$ of $M^+(g)$ is 150 eV

3) $I.E_2$ of $M(g)$ is 250 eV 4) $I.E_2$ of $M(g)$ is 150 eV

22. Some statements are given. Among them the correct statements are

I) Second ionization potential of sodium is greater than that of Magnesium

II) Second ionization potential of lithium greater than first ionization potential of Helium

III) Second ionization potential of sodium greater than first ionization potential of Neon

IV) First ionization potential of oxygen is greater than that of Nitrogen

1) Only I, II and III are correct 2) Only I, II are correct

3) Only I, IV are correct 4) All are correct

23. The amount of energy released when one million atoms of gaseous iodine are completely converted into $I^-(g)$ is $4.825\times10^{-13} J$. Calculate the electron gain enthalpy of iodine in eV/atom ? **(given 1eV/atom = 96.5 kJ/mole)**

24. Identify the correct statement?

1) $B<C<N<O$: First ionization enthalpy

2) $S^{2-}>Cl^->K^+>Ca^{2+}$: Ionization enthalpy

3) $Al^{3+}<Mg^{2+}<Na^+<F^-$: Ionic size

4) $Cu>Ag>Au$: First ionization enthalpy

25. First and second ionization energy of an element M are 600 kJ / mol and 1600 kJ/mol respectively. If 1000kJ of energy is supplied to one mole of vapours of M then in the resulting mixture mole percentage of M^{2+} is

26. Which one of the following statement is incorrect?

1) Greater the nuclear charge, more negative is the electron gain enthalpy

2) Nitrogen has positive electron gain enthalpy

3) Electron gain enthalpy decreases from fluorine to iodine in the group

4) Chlorine has highest electron gain enthalpy with negative sign.

27. Which of the following is exothermic step?

1) $Na^+(g)+Cl^-(g)\rightarrow NaCl(s)$ 2) $S^-(g)\rightarrow S^{2-}(g)$

3) $N(g)\rightarrow N^-(g)$ 4) $F^-(g)\rightarrow F(g)$

28. Consider the following orders:

i) $N_2O_3>N_2O_4>N_2O_5$: Acidic character

ii) $CH_4<CCl_4<CF_4$: Electronegativity of central 'C'-atom

iii) $Mg^{2+}<K^+<S^{2-}<Se^{2-}$: Ionic radius

iv) $Ni>Pd>Pt$: Ionisation energy

v) $As^{5+}>Sb^{5+}>Bi^{5+}$: Stability of oxidation state

vi) $LiF>NaF>KF>RbF$: Lattice energy

vii) $Cl>Br>F>I$: Electron affinity

Then calculate the value of $|x-y|$, where 'x' and 'y' are correct and incorrect orders respectively.

29. The electronic configuration of 4 elements are :

(I) $[Kr]5s^1$ (II) $[Rn]5f^{14}6d^1 7s^2$ (III) $[Ar]3d^{10}4s^2 4p^5$ (III) $[Ar]3d^6 4s^2$

Consider the following statements:

i) I shows variable oxidation state

ii) II is a d – block element.

iii) The compound formed between I and III is covalent.

iv) IV shows single oxidation state.

Which of the following statements is True (T) or False (F)

1) F T F F 2) F T F T 3) F F T F 4) F F F F

30. Which of the following is incorrect?

1) $Ar-K$; $Te-I$; $Co-Ni$ are anomalous pairs present in Mendeleev's periodic table.

2) Cu^+ and Zn^{2+} have Pseudo inert gas configuration.

3) The maximum covalence of boron is 4 where as that of Al is 6

4) increasing value of first ΔH_{eg} (magnitude) : O < S < Se

CHEMICAL BOND

31. Correct order of covalent nature is

1) $CsCl > KCl > NaCl$ 2) $SnCl_4 > SiCl_4 > CCl_4$

3) $ZnCl_2 > CdCl_2 > HgCl_2$ 4) $HgCl_2 > ZnCl_2 > CdCl_2$

32. How many of the following molecules have expanded octet in the valence shell of the central atom ? $PCl_5, CCl_4, BF_3, SF_6, XeF_6, IF_7, BeF_2, H_2O, BrF_5, NH_3$

33. Which of the following is incorrect?

1) order of lattice energy : LiF > NaF > KF > RbF > CsF

2) order of covalent character : $CaI_2 < CaF_2$

3) order of M.P : $NaF > NaCl > NaBr > NaI$

4) order of thermal stability : $MgCO_3 < CaCO_3 < SrCO_3 < BaCO_3$

34. Select the IN**CORRECT** statement.

1) Polarizability of S^{2-} is lower than Se^{2-}
2) Polarizability of S^{2-} is higher than Cl^{-}
3) Polarizability of S^{2-} is lower than P^{3-}
4) Polarizability of O^{2-} is higher than S^{2-}

35. In which there is change in the type of hybridisation when :

1) NH_3 combines with H^+
2) AlH_3 combines with H^-
3) NH_3 forms NH_2^-
4) H_2O forms H_3O^+

36. The shapes of PCl_4^+, PCl_4^- and $AsCl_5$ are respectively:

1) square planar, tetrahedral and see-saw
2) tetrahedral, see-saw and trigonal bipyramidal
3) tetrahedral, square planar and pentagonal bipyramidal
4) trigonal bipyramidal, tetrahedral and square pyramidal

37. Among the molecules $CH_4, PCl_5(g), PCl_3F_2, ClF_3, CO_2, SF_4, SF_2Cl_2, CCl_4, SCl_4, and\ H_2O.$, the number of molecules having different bond lengths are

38. Match List I with List II:

List-I (molecules/ions)		List-II (No. of lone pairs of e^- on central atom)	
a	IF_7	I	Three
b	ICl_4^-	II	One
c	XeF_6	III	Two
d	XeF_2	IV	Zero

Choose the **Correct** answer from the options given below:

1) a – II, b – III, c – IV, d – I
1) a – IV, b – III, c – II, d – I
3) a – II, b – I, c – IV, d – III
4) a – IV, b – I, c – II, d – III

39. Identify the incorrect statement in the following:

1) Hybridisation of carbon in $C_3O_2\, is\, SP$

2) $N(Me)_3$ and $N(SiMe_3)_3$ are iso structural

3) In trigonal bipyramidal arrangement lone pair of electron are usually occupied at equatorial position.

4) The shape of I_3^- is linear.

40. The difference between bond orders of CO and NO^+ is $\frac{x}{2}$ where $x =$ _____

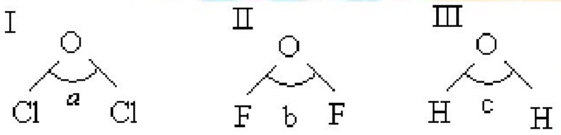
(Round off to the nearest Integer)

41. The correct order of increasing intermolecular hydrogen bond strength is

1) $HCN < H_2O < NH_3$ 2) $HCN < CH_4 < NH_3$

3) $CH_4 < HCN < NH_3$ 4) $CH_4 < NH_3 < HCN$

42. In the following compounds:-



Which is the correct order for bond angle?

1) $a > b > c$ 2) $b > c > a$ 3) $a > c > b$ 4) $b < a < c$

43.

Oxides of sulphur	Number of $(p\pi - p\pi)$ bonds	Number of $(d\pi - p\pi)$ bonds
SO_2	**x**	**a**
SO_3	**y**	**b**

Value of $\frac{(x+y)}{(a \times b)}$ is ………

(Given that: consider expanded octet structures for the given oxides)

44. According to MO theory the bond orders for O_2^{2-}, CO and NO^+ respectively, are

1) 1, 3 and 3 2) 1, 3 and 2 3) 1, 2 and 3 4) 2, 3 and 3

45. Increasing order of carbon-oxygen bond length in CO, CO_2 and CO_3^{2-} is

1) $CO < CO_2 < CO_3^{2-}$
2) $CO < CO_3^{2-} < CO_2$
3) $CO_3^{2-} < CO_2 < CO$
4) $CO_3^{2-} < CO_2 = CO$

46. Molecule AB has a bond length of 1.617A and a dipole moment of 0.38 D. The fractional charge on each atom (absolute magnitude) is : $\left(e_0 = 4.802 \times 10^{-10} esu\right)$

1. 0 2. 0.05 3. 0.5 4. 1.0

47. Ice made from heavy water is put in container A having normal water and ice made from normal water is put in container B containing heavy water.
The observation are:
1) Ice sinks in both container A and B
2) Ice floats in both container A and B
3) Ice sinks in container A and floats in container B.
4) Ice floats in container A and sinks in container B.

48. In which of the following transformations both HOMO and magnetic behavior changes? I) $O_2^+ \rightarrow O_2^{2+}$ II) $N_2 \rightarrow N_2^{2-}$ III) $O_2^{2+} \rightarrow O_2$ IV) $C_2 \rightarrow C_2^+$

1) I, II, IV 2) II, III, IV 3) I, II, III 4) I,II, III, IV

49. Which among the following statements are correct?

a) order of 'O – O' bond length is $O_2\left[AsF_6\right] < O_2 < KO_2$

b) Though ethyl alcohol and dimethyl ether have same molecular weights, dimethyl ether is more volatile then ethyl alcohol.

c) the order of bond order is : $N_2 > N_2{}^+ > N_2{}^{2-}$

d) ice has more density than water

1) a, b, d 2) a, b, c 3) a, c, d 4) a, b, c, d

50. Which of the following statements is true?

1) the dipole moment of NF_3 is more than NH_3

2) trans – 2, 3 – dichloro – 2 – pentene does not have net permanent dipole moment

3) Carbon tetrachloride has no net dipole moment because of its regular tetrahedral structure

4) Dipole moment is zero for sulphur dioxide molecule

S – BLOCK ELEMENTS

51. Which is not correctly matched?

A) Basic strength of oxides $Cs_2O < Rb_2O < K_2O < Na_2O < Li_2O$

B) Stability of peroxides $Na_2O_2 < K_2O_2 < Rb_2O_2 < Cs_2O_2$

C) Stability of bicarbonate $LiHCO_3 < NaHCO_3 < KHCO_3 < RbHCO_3$

D) Melting point $NaF < NaCl < NaBr < NaI$

1) A and D 2) A and C 3) A and B 4) B and C

52. Property of all the alkali metals that decrease with their atomic number is/are :

I) Solubility of their hydroxides

II) Thermal stability of their carbonates

III) Softness IV) Hydration energy

1) III, IV 2) II, IV 3) I only 4) IV only

53. Given below are two statements:

Assertion (A): The alkali metals and their salts impart characteristic colour to reducing flame.

Reason (R): Alkali metals can be detected using flame tests.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both A and R are correct but R is not the correct explanation of A.

2) A is correct but R is not correct.

3) A is not correct but R is correct.

4) Both A and R are correct and R is the correct explanation of A.

54. **Statement-1:** LiF and CsI have low solubility in water.

Statement-2: Both have high lattice enthalpy

1) Statement-1 is True,Statement-2 is True; Statement-2 is a correct explanation for Statement1.

2) Statement-1 is True, Statement-2 is True, Statement-2 is NOT a correct explanation

for Statement-1.

3) Statement-1 is True, Statement-2 is False.

4) Statement-1 is False, Statement-2 is True.

55. Metal M + air $\xrightarrow{\Delta}$ A $\xrightarrow{H_2O}$ B $\xrightarrow{HCl}$ white fumes; Metal M can be:

1) Li, Mg 2) Li, Al or K

3) Na, K or Mg 4) Li, Na or K

56. The incorrect statements among the following are

1) In the reaction of NaOH with 'CO' at $150^0C - 200^0C$ and $5-10\,atm.$ pressure, CO acts as Lewis acid.

2) $BeCO_3$ to be stored in atmosphere of CO_2

3) Beryllium fluoride is highly soluble in water, while CaF_2 is almost insoluble

4) Metals can be recovered on evaporating ammonia from the solution of Alkaline earth metals in liquid ammonia.

57. The reaction of an element A with water produces combustible gas B and an aqueous solution of C. When another substance D reacts with this solution c also produces the same gas B. D also produces the same gas even on reaction with dilute H_2SO_4 at room temperature. Element A imparts golden yellow colour to Bunsen flame. Then, A, B, C and D may be identified as:

1) Na, H_2, NaOH and Zn 2) K, H_2, KOH and Zn

3) K, H_2, NaOH and Zn 4) Ca, H_2, $CaCOH_2$ and Zn

58. Which is incorrect about highly pure dilute solution of sodium in liquid ammonia :

1) On evaporation yield metals 2) Exhibits electrical conductivity

3) Produces sodium amide and hydrogen gas instantly

4) Acts as powerful reducing agent

59. When milk of lime is added to an aqueous solution of 'A'. $CaCO_3$ is precipitated along with an aqueous solution of 'B'. When CO_2 is passed into B (not excess) again A is formed. A and B

1) $NaOH$ and $NaHCO_3$ 2) $NaHCO_3$ and Na_2CO_3

3) $NaOH$ and Na_2CO_3 4) Na_2CO_3 and $NaOH$

60. The wrong statement among the following regarding alkali metal halides is

1) ΔH_f^0 values becomes less negative from lithium to caesium

2) For a given alkali metal $\Delta f H^0$ is in the order

fluoride > chloride > bromide > iodide

3) When saturated aqueous solution of LiI and CsF are mixed, LiF and CsI are crystallized

4) Lithium halides have more melting points than sodium halides

61. The incorrect statement among the following is:

1) CaO violently reacts with water

2) Gypsum on heating at 393 K gives calcium sulphate hemihydrate

3) Sodium amalgam reacts with water and give NaOH with liberation of H_2

4) KO_2 absorbs CO_2 and releases CO gas

62. Which metal bicarbonates does not exist in solid state?

(i) $LiHCO_3$ (ii) $Ca(HCO_3)_2$ (iii) $Zn(HCO_3)_2$ (iv) $AgHCO_3$

1) i, ii, iii, iv 2) i, ii, iii 3) i, ii, iv 4) ii, iii, iv

63. How many of the following produce $H_{2(g)}$ on reaction with NaOH –

B, Si , Al, Zn, Cl_2, $SnCl_2$, ,HNO_3, PbO_2, H_2S

64. Which salt reacts with NaOH to give white precipitate which is soluble in excess NaOH ?

I) $ZnCl_2$ II) $AlCl_3$ III) $CuSO_4$ IV) $FeSO_4$

1) I, II 2) II, III 3) III, IV 4) II,III,IV

65. Which of the following is not the use of alkali metals?

1) White metal, an alloy of lithium and lead is used in manufacture of bearings for motor engines

2) Sodium – lead alloy is used in the preparation tetraethyl lead an anti

knocking agent

3) Potassium chloride is used as fertilizer

4) The alkali metals used in photoelectric cells are lithium and sodium

66.

Column-I		Column-II	
(a)	CaH_2	(p)	Paramagnetic anion
(b)	K_2O_2	(q)	Homo diatomic, diamagnetic anion
(c)	KO_2	(r)	Neutral aqueous solution
(d)	NaCl	(s)	Gives hydrogen on hydrolysis

(1) $a \to s; b \to q; c \to p; d \to r$ (2) $a \to s; b \to q; c \to r; d \to p$

(3) $a \to s; b \to r; c \to p; d \to q$ (4) $a \to p; b \to q; c \to s; d \to r$

67. Which of the following statement is wrong?

1) Though sodium and potassium are similar chemically, they differ in their behavior in biological system

2) Potassium ions are less abundant within the cell than sodium ions as their penetration power is less

3) Potassium ions activate many enzymes

4) Sodium ions are responsible for the transmission of nerve signals

68. The alkaline earth metal sulphate(s) which are readily soluble in water is/are:

(a) $BeSO_4$ (b) $MgSO_4$ (c) $CaSO_4$ (d) $SrSO_4$ (e) $BaSO_4$

Choose the correct answer from the options given below:

1) a only 2) b only 3) a and b 4) b and c

69. How many of the following metals can be detected in the laboratory through flame test

i) Be ii)Mg iii) Na iv) K v)Ba

70. Select the incorrect statement(s) :

(1) Magnesium can be burnt in the atmosphere of CO_2 and SO_2

(2) Magnesium reacts with alkyl halides to form Grignard's reagent

(3) Out of Mg and Ca, only Mg reacts with N_2 to form magnesium nitride

(4) BeO is amphoteric in nature.

71. IIA Group halides which doesn't follow the solubility order in water as

$MF_2 < MCl_2 < MBr_2 < MI_2$,

Find its atomic number of that IIA group element

1) 4 2) 11 3) 19 4) 4 or 11 or 19

72. How many of the following metals whose nitrate salts decompose to give mixture of $NO_{2(g)}$ and $O_{2(g)}$

i) Li ii) Na iii) Ca iv) K v) Mg vi) Sr vii) Ba

73. Identify the incorrect statement

1) Beryllium chloride is hygroscopic and fumes in moist air due to hydrolysis

2) Anhydrous $BeCl_2$ can be obtained by heating $\left[Be(H_2O)_4\right]Cl_2$

3) $BeCl_2$ in solid state is polymeric and contain $(3c-4e)$ bridge bonds.

4) Beryllium fluoride is highly soluble in water where as CaF_2 is insoluble

74. EDTA is used in the estimation of:

(1) Mg^{2+} ions (2) Ca^{2+} ions

(3) both Ca^{2+} and Mg^{2+} ions (4) Mg^{2+} ions but not Ca^{2+} ions

75. Amongst the following hydroxides, the one which has the highest value of K_{sp} at ordinary temperature is:

(1) $Mg(OH)_2$ (2) $Ca(OH)_2$ (3) $Sr(OH)_2$ (4) $Ba(OH)_2$

IIIA GROUP ELEMENTS

76. Identify incorrect order

1) Atomic radius: $B < Ga < Al < In < Tl$

2) Ionisation enthalpy (IP_1) : $In < Al < Ga < Tl < B$

3) Electro negativity: $B > Al > In > Ga > Tl$

4) Order of stability : $Tl^{+1} > Tl^{+3}$

77. Al and Ga have almost same covalent radius because of

1) greater shielding power of s-electrons of Ga atom

2) poor shielding power of s-electrons of Ga atom

3) poor shielding power of d-electrons of Ga atom

4) greater shielding power of d-electrons of Ga atom

78. In the following sets of reactants which two sets best exhibit the amphoteric character of $Al_2O_3.xH_2O$?

Set 1: $Al_2O_3.xH_2O(s)$ and OH^- (aq); Set 2: $Al_2O_3.xH_2O(s)$ and $H_2O(l)$

Set 3: $Al_2O_3.xH_2O(s)$ and $H^+(aq)$ Set 4: $Al_2O_3.xH_2O(s)$ and $NH_3(aq)$

1) 1 and 2 2) 1 and 3 3) 2 and 4 4) 3 and 4

79. Assertion (A): Among III A group elements gallium has lowest melting point.

Reason (R): Gallium exists as giant covalent polymer in both liquid and solid states

1) A is true and R is false 2) A is false and R is true

3) Both A and R are true. R Explains A

4) Both A and R are true. R does not explain A

80.

Crystalline Borax (Molecular structure)		**Sodium peroxoborate (Molecular Structure)**	
(i)	Number of $B-O-B$ bonds is 'x'	**(i)**	Number of $B-O-B$ bonds is 'a'
(ii)	Number of $B-B$ bonds is 'y'	**(ii)**	Number of $B-O-O-B$ bonds is 'b'
(iii)	Number of $sp2$ hybridized Boron atoms is 'z'	**(iii)**	Number of $sp3$ hybridized Boron atoms is 'c'

The value of $[(x+y+z)-(a+b+c)]$ is ________

81. Three moles of B_2H_6 are completely reacted with methanol. The number of moles of boron containing product formed is

82. Which of the following is/are cleavage reaction of Diborane?

i) $B_2H_6 + Cl_2 \rightarrow$ ii) $B_2H_6 + 2CO \rightarrow$ iii) $B_2H_6 + 2NMe_3 \rightarrow$

iv) $B_2H_6 + 2NH_3 \xrightarrow{lowtemperature}$

1) i, ii and iii 2) i, iii and iv 3) i and iii 4) i, ii, iii, iv

83. $Na_2B_4O_7.10H_2O \xrightarrow{heat} X + NaBO_2 + H_2O,$

$X + Cr_2O_3 \xrightarrow{heat} Y\,(green\,coloured)$ X and Y are:

1) Na_3BO_3 and $Cr(BO_2)_3$ 2) $Na_2B_4O_7$ and $Cr(BO_2)_3$

3) B_2O_3 and $Cr(BO_2)$ 4) B_2O_3 and $Cr(BO_2)_3$

84. The compound added to the following reaction to make it proceed in forward direction

$H_3BO_3 + H_2O \rightleftharpoons [B(OH)_4]^- + H^+$

1) Addition of Borax 2) Addition of trans 1, 2 diol

3) Addition of cis 1, 2 diol 4) Addition of Na_2HPO_4

85. The structure of diborane contains

1) Four 2c-2e bonds and four 3c-2e bonds

2) Two 2c-2e bonds and two 3c-2e bonds

3) Two 2c-2e bonds and four 3c-2e bonds

4) Four 2c-2e bonds and two 3c-2e bonds

86. $\mathrm{NaBH_4 + I_2 \longrightarrow NaI + X\uparrow (diatomic) + A\uparrow}$

incorrect statements regarding ‘A’ is:

1) ‘A’ catches fire spontaneously upon exposure to air

2) One mole of ‘A’ on hydrolysis with excess water gives 6 moles of gas ‘X’

3) On an industrial scale ‘A’ can be produced by the reaction of $\mathrm{BF_3}$ with NaH on heating at 450K

4) ‘A’ undergo unsymmetrical cleavage with CO.

87. In which of the following reactions H_2 gas is not liberated.

1) $B_2H_6 + H_2O \rightarrow$ 2) $B_2H_6 + KOH + H_2O \rightarrow$

3) $B_2H_6 + Br_2 \xrightarrow{100^0C}$ 4) $B_2H_6 + HCl \xrightarrow{AlCl_3}$

88. $B_2H_6 + H_2O \xrightarrow{low\ temperature} Z + H_2$,

$Z \xrightarrow{>370k} X \xrightarrow{\Delta,\text{Red hot}} Y$; X and Y respectively are

1) X = Meta boric acid ; Y = Boric oxide.

2) X = Borax; Y = Meta boric acid

3) X = Tetra boric acid; Y = Meta boric acid

4) X = Meta boric acid Y = Tetra boric acid Borax

89. How many of the following statements are **correct** regarding borax ?

(i) It consist of two six membered rings

(ii) It consists of all atoms in same plane

(iii) It is used as primary standard for titrating acids

(iv) Each boron is attached to one hydroxide group

(v) On heating it forms glassy transparent residue consist of Na_3BO_3 and B_2O_3

90. In gold Schmidt aluminothermic process which of the following reducing agents is used:

1) Coke 2) Al-powder 3) calcium 4) sodium

91. Which of the following is an electron deficient molecule?

1) LiH 2) $(AlH_3)_n$ 3) $LiBH_4$ 4) $B_3N_3H_6$

92. Be and Al show diagonal relationship and thus

1) Their oxides are soluble in alkali solution forming $[Be(OH)_4]^{+2}$, $[Al(OH)_6]^{3-}$

2) They form complex anion BeF_4^{2-} and AlF_6^{3-}

3) $BeCl_2$ is Lewis acid but $AlCl_3$ is Lewis base

4) BeO basic but Al_2O_3 is amphoteric

93. How many of the following consist of sp^2 hybridized atom(s) ?

Diamond, BN, $Na_2B_4O_7 \cdot 10H_2O$, BF_3, graphite, Silicon(s), Al_2Cl_6, $Al_2(CH_3)_6$.

94. How any of the following reacts with both acids and bases?

$Al_2O_3, BeO, B_2O_3, Al, Ga_2O_3, Tl_2O$

95. Which of the the following combination doesnot lead to formation of anhydrous $AlCl_3$?

1) $AlCl_3 \cdot 6H_2O$, Heat
2) Al, dry HCl, Heat
3) $AlCl_3 \cdot 6H_2O, SOCl_2$, Heat
4) Al_2O_3, C, Cl_2, Heat

IVA GROUP ELEMENTS

96. The correct order of bond enthalpy (kJ mol^{-1}) is____.

1) $Si-Si > C-C > Sn-Sn > Ge-Ge$
2) $Si-Si > C-C > Ge-Ge > Sn-Sn$
3) $C-C > Si-Si > Sn-Sn > Ge-Ge$
4) $C-C > Si-Si > Ge-Ge > Sn-Sn$

97. Which of the following halides does not exist?

(1) PbF_4 (2) $PbCl_4$ (3) $PbBr_4$ (4) PbI_4

98. In C_{60}, if a five membered ring is fused with 'x' six membered and 'y' membered rings. Find the value of $(x+y)$?

99. Water gas is prepared by

1) passing air through red hot coke
2) passing steam through white hot coke
3) passing air and steam through red hot coke
4) none of these

100. Regarding diamond

1) C – C bond length is 1.54A°
2) It has least refractive index among solids
3) It is a 3–D polymer

The correct combination is

1) all are correct
2) I & III are correct
3) I & II are correct
4) II & III are correct

101. (A): CCl_4 does not undergo hydorlysis whereas $SiCl_4$ is readily hydrolysed.

(R): Carbon has no vacant d - orbitals, but silicon has vacant d - orbitals.

The correct answer is

1. Both A and R are true and R is the correct explanation of A.
2. Both A and R are true and R is not the correct explanation of A.
3. A is true but R is false

4. A is false but R is true.

102. Select the incorrect statement

1) CO_2 is more acidic than SiO_2

2) A hydrate $CO_2.8H_2O$ can be crystallised.

3) CO is neutral gas while PbO is acidic

4) Carbonic acid never been isolated, only exist in the solution.

103. Which of the following statement is incorrect regarding the compounds of carbon family elements?

1) Maximum coordination number of carbon in commonly occurring compounds is 4, where as that of silicon is 6.

2) The stability order of dihalides is $SiX_2 < GeX_2 < SnX_2 < PbX_2$.

3) The order of boiling point of hydrides of is $CH_4 < SiH_4 < GeH_4 < SnH_4$.

4) $MeSiCl_3$ on hydrolysis and subsequent condensation will produce $(Me)Si(OH)_3$.

104. Which of the following is false about silicones?

1) They are formed by hydrolysis of $R_2\ SiCl_2$

2) They are polymer, made up of $R_2\ SiO_2$ units

3) They are made up of SiO_4^{4-} units

4) They are macromolecules

105. SiO_2 reacts with

(a) Na_2CO_3 (b) CO_2 (c) HF (d) HC*l*

1) a,c 2) b,c 3) c,d 4) a,d

106. Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Carbon forms two important oxides CO and CO_2. CO is neutral whereas CO_2 is acidic in nature.

Reason R : CO_2 can combine with water in a limited way to form carbonic acid, while CO is sparingly soluble in water.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both A and R are correct but R is **NOT** the correct explanation of A.

2) Both A and R are correct and R is the correct explanation of A.

3) A is not correct but R is correct.

4) A is correct but R is not correct.

107. A mixture of two gases (x and y) is obtained by the passage of steam over hot coke at 473 – 1273K. Then identify the **TRUE/FALSE** statement(s) regarding the gaseous products(x and y) ?

I) One of the gaseous product is used as a reducing agent for the extraction of Al from Al_2O_3 (commercial scale)

II) One of the gaseous product is estimated by using I_2O_5

III) One of the gaseous product is used for the manufacture of ammonia by the Haber process

IV) All the elements of Group 2 readily combine with one of the gaseous product upon heating to form their hydrides (MH_2) o

1) T, F, F, T 2) F, T, T, F 3) F, F, T, T 4) T, T, T, T

108. Formula of a single chain silicate (pyroxene) containing three silicate units is $Si_3O_x^{m-}$ and formula of a cyclic silicate containing three silicate units is $Si_3O_y^{n-}$, then the value of $\left[(x+y)-(m+n)\right]$ is

109. Which of the following is a good oxidising agent?

1) C^{4+} 2) Pb^{4+} 3) Sn^{4+} 4) Ge^{4+}

ENVIRONMENTAL CHEMISTRY

110. The correct set of species responsible for the photochemical smog is

1) N_2, O_2, O_3 and hydrocarbons 2) N_2, NO_2 and hydrocarbons

3) CO_2, NO_2, SO_2 and hydrocarbons 4) NO, NO_2, O_3 and hydrocarbons

111. The correct set of species responsible for the photochemical smog is

1) N_2, O_2, O_3 and hydrocarbons 2) N_2, NO_2 and hydrocarbons

3) CO_2, NO_2, SO_2 and hydrocarbons 4) NO, NO_2, O_3 and hydrocarbons

112. Which of the following statements is incorrect?

1) Higher concentration of NO_2 damages the leaves of plants and retard the rate of photosynthesis

2) When the pH of the rain water drops below 5.6, it is called acid rain.

3) The fluoride ions make the enamel on teeth much harder by converting hydroxyapatite, the enamel on the surface of the teeth, into much harder chlorapatite.

4) Excess sulphate (> 500ppm) in drinking water causes laxative effect.

113. Correct statement is :

1) An average human being consumers more food than air

2) An average human being consumes nearly 15 times more air than food

3) An average human being consumes equal amount of food and air

4) An average human being consumes 100 times more air than food

114. Formation of photochemical smog involves the following reaction in which A, B and C are respectively.

I) $NO_2 \xrightarrow{h\nu} A+B$ II) $B + O_2 \rightarrow C$ III) $A + C \rightarrow NO_2 + O_2$

Choose the correct answer from the options given below :

1) O, NO & NO_3^- 2) O, N_2O & NO 3) N, O_2 & O_3 4) NO, O & O_3

115. The industrial activity held least responsible for global warming is :

1) Manufacturing of cement 2) Steel manufacturing

3) Electricity generation in thermal power plants

4) Industrial production of urea

116. The eutrophication of water body results in:

1) loss of biodiversity 2) breakdown of organic matter

3) increase in Biodiversity 4) decrease in BOD

117. Which among the following is not a pesticide?

1) DDT 2) Organophosphates 3) Dieldrin 4) Sodium arsenite

118. Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Polluted water may have a value of BOD of the order of 17 ppm.

Reason R : BOD is a measure of oxygen required to oxidise both the biodegradable and non-biodegradable organic material in water.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both A and R are correct but R is **NOT** the correct explanation of A.

2) Both A and R are correct and R is the correct explanation of A.

3) A is not correct but R is correct.

4) A is correct but R is not correct.

KEY

1	**4**	2	**4**	3	**1**	4	**1**	5	**400**
6	**2**	7	**8**	8	**4**	9	**1**	10	**3**
11	**1**	12	**4**	13	**8**	14	**1**	15	**4**
16	**2**	17	**2**	18	**1**	19	**1**	20	**3**
21	**1,3**	22	**1**	23	**3**	24	**3**	25	**25**
26	**3**	27	**1**	28	**1**	29	**4**	30	**4**
31	**4**	32	**5**	33	**2**	34	**4**	35	**2**
36	**2**	37	**6**	38	**2**	39	**2**	40	**0**
41	**3**	42	**3**	43	**1**	44	**1**	45	**1**
46	**2**	47	**3**	48	**3**	49	**2**	50	**3**
51	**1**	52	**4**	53	**3**	54	**3**	55	**1**
56	**4**	57	**1**	58	**3**	59	**4**	60	**4**
61	**4**	62	**1**	63	**4**	64	**1**	65	**4**
66	**1**	67	**2**	68	**3**	69	**3**	70	**3**
71	**1**	72	**5**	73	**2**	74	**3**	75	**4**
76	**3**	77	**3**	78	**2**	79	**1**	80	**3**
81	**6**	82	**4**	83	**4**	84	**3**	85	**4**
86	**4**	87	**3**	88	**1**	89	**3**	90	**2**
91	**2**	92	**2**	93	**4**	94	**5**	95	**1**
96	**4**	97	**4**	98	**5**	99	**2**	100	**2**
101	**1**	102	**3**	103	**4**	104	**3**	105	**1**
106	**2**	107	**2**	108	**5**	109	**2**	110	**4**
111	**4**	112	**3**	113	**2**	114	**4**	115	**4**
116	**1**	117	**4**	118	**4**				

HINTS

5. **Hint:** m. eqts. of Na_2CO_3 used $=(100-60)\times\frac{1}{50}=0.8\,m.eqts.$

m. eqts. of Na_2CO_3 used = m. eqts. of Ca^{2+} or Mg^{2+} = m. eqts. of $CaCO_3$ = 0.8

mass of $CaCO_3$ $=0.8\times50=40\,mg$

100 g of water $\equiv40\times10^{-3}\,g$

10^6 *g of water* $\equiv\frac{40\times10^{-3}}{100}\times10^6=400$ *ppm of* $CaCO_3$

7. **Hint:** $PbS+4H_2O_2\rightarrow PbSO_4+4H_2O$

14. **Hint:** $Zn+2NaOH\rightarrow Na_2ZnO_2+H_2$ (NCERT)

$Zn+2NaOH+2H_2O\rightarrow Na_2\left[Zn(OH)_4\right]+H_2$

15. **Hint**: Dielectric constant and degree of dissociation is less for D_2O than H_2O

23. **Hint:** The energy released for 1 mole of iodine atoms $=\frac{4.825\times10^{-13}\times6\times10^{23}}{10^6}=289.5\,kJ$

The electron gain enthalpy of iodine $=\frac{289.5}{96.5}=3eV/atom$

25. **Hint:** $M\rightarrow M^++e^-$; $M^+\rightarrow M^{2+}+e^-$

$1000-600=400$ (used for second ionization)

$\frac{400}{1600}=0.25$ moles of M^{2+}, therefore mole % = 25

37. **Hint:** In Trigonal bipyramid geometry axial bonds are longer than equatorial bonds. Thus, $PCl_5, PCl_3F_2, SF_4, SF_2Cl_2, SCl_4, ClF_3$ have different bond lengths.

43. Hint: SO_2: 2 π bonds – one $(p\pi-p\pi)$ **and one** $(d\pi-p\pi)$

SO_3 : 3 π bonds – one $(p\pi-p\pi)$ **and two** $(d\pi-p\pi)$

81. **Hint:** $B_2H_6+6CH_3OH\rightarrow2B(OCH_3)_3+6H_2$

For 3 moles of B_2H_6, moles of 'B' containing product formed = 6

86. **HINT**: $2NaBH_4+I_2\rightarrow B_2H_6+2NaI+H_2$

$2BH_3+6NaOH\xrightarrow{450K}B_2H_6+2NaF$

$B_2H_6+2CO\longrightarrow2[H_3B\leftarrow CO]$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